THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 1: **Orienting Yourself: The Use of Coordinates** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The point $(-3, 5)$ lies: 1

(a) Quadrant III **(b)** Quadrant II **(c)** Quadrant IV **(d)** Quadrant I

2. The point $(0, -4)$ lies: 1

(a) on the y-axis **(b)** Quadrant II

(c) Quadrant III **(d)** Quadrant I

3. The distance between $A(0, 0)$ and $B(6, 8)$ is: 1

(a) 9 units **(b)** 10 units **(c)** 11 units **(d)** 12 units

4. The distance between $A(-1, -1)$ and $B(4, 11)$ is: 1

(a) 13 units **(b)** 12 units **(c)** 15 units **(d)** 14 units

5. The midpoint of the segment joining $A(5, 9)$ and $B(-1, -3)$ is: 1

(a) $(-3, -6)$ **(b)** $(2, -3)$ **(c)** $(2, 3)$ **(d)** $(4, 6)$

6. $M(0, 0)$ is the midpoint of AB and $A(-3, 7)$. The coordinates of B are: 1

(a) $(3, -7)$ **(b)** $(4, -6)$ **(c)** $(3, 7)$ **(d)** $(-3, 7)$

7. The distance of the point $(-8, -15)$ from the origin is: 1

(a) 18 units **(b)** 8 units **(c)** 23 units **(d)** 17 units

8. The reflection of the point $(-8, -1)$ in the y-axis is: 1

(a) $(-8, 1)$ **(b)** $(8, -1)$ **(c)** $(-1, -8)$ **(d)** $(8, 1)$

9. The point (a, b) coincides with the point (b, a) if and only if: 1

(a) $a = -b$ **(b)** $a = b$ **(c)** $a + b = 0$ **(d)** $a = 0$

10. A student says: "The points $(2, 3)$ and $(3, 2)$ are the same point." The correct response is: 1

(a) True if they are on the same line

(b) True only in Quadrant I

(c) True

(d) False; they are different because $(x, y) \neq (y, x)$ when $x \neq y$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the distance between $A(2, 3)$ and $B(5, 7)$. 2

12. Find the distance between $A(-1, 2)$ and $B(4, -10)$. 2

13. Find the distance between $A(0,\ 5)$ and $B(12,\ 0)$. 2

14. Find the distance between $A(-3,\ -3)$ and $B(5,\ 3)$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Show that the triangle with vertices $A(0,\ 0)$, $B(6,\ 0)$, $C(3,\ 4)$ is isosceles. 3

16. Show that the triangle with vertices $A(1,\ 1)$, $B(7,\ 1)$, $C(4,\ 5)$ is isosceles. 3

17. Show that the triangle with vertices $A(-2,\ 0)$, $B(4,\ 0)$, $C(1,\ 4)$ is isosceles. 3

OR

Show that the triangle with vertices $A(2,\ 3)$, $B(8,\ 3)$, $C(5,\ 7)$ is isosceles. 3

18. Show that $A(0,\ 0)$, $B(4,\ 0)$, $C(4,\ 3)$, $D(0,\ 3)$ are the vertices of a rectangle. 3

OR

Show that $A(1,\ 1)$, $B(5,\ 1)$, $C(5,\ 4)$, $D(1,\ 4)$ are the vertices of a rectangle. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A city has two main roads crossing at the centre, N–S and E–W. All other streets are parallel and 200 m apart, with 10 streets in each direction. A street intersection is named (a, b) if the a-th N–S street meets the b-th E–W street. 5

(i) How many intersections are there in all?

(ii) How far apart are the crossings $(2, 3)$ and $(6, 3)$?

(iii) Find the straight-line distance between $(2, 3)$ and $(6, 6)$ in metres.

OR

(i) Find the distance between $P(-2,\ 3)$ and $Q(4,\ -5)$. 5

(ii) Hence find the midpoint of PQ and verify that it is equidistant from P and Q.

20. Show that the points $A(1, 7)$, $B(4, 2)$, $C(-1, -1)$ and $D(-4, 4)$ are the vertices of a square. 5

OR

Case study: On a grid map of a school, the library is at $L(2,\ 3)$, the canteen at $C(8,\ 3)$ and the playground gate at $G(5,\ 7)$ (units in 10 m). 5

(i) Show that $LG = CG$.

(ii) Find the midpoint of LC (the shaded sitting area).

(iii) Find the distance from G to this midpoint and state what it represents.

END OF QUESTION PAPER — Total: 40 Marks

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